

The Energy Trail

II PUC · Ch 12 · Ecosystem

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



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The Pond Village

"An **ecosystem** is a living community together with its non-living surroundings," said Prof. BAIO by a pond. "**Producers** make food, **consumers** eat it and **decomposers** recycle the dead." In a forest, plants grow in layers — **stratification**.

💡 REMEMBER

A pond is a small, self-sustaining ecosystem.



Making Food

Plants make organic matter by photosynthesis — **gross primary productivity** (GPP). After their own respiration, what is left for others is **net primary productivity** (NPP = GPP − R). The whole biosphere makes about 170 billion tons of dry matter a year.

💡 **REMEMBER**

Oceans cover 70% of Earth but make only about 55 billion tons.



The Clean-up Crew

Decomposition happens in steps: **fragmentation** by detritivores like earthworms, **leaching** of nutrients into the soil, **catabolism** by enzymes, **humification** into dark **humus** and **mineralisation** into inorganic nutrients.

💡 **REMEMBER**

Decomposition is fast when warm, moist and aerobic; slow for lignin and chitin.



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Follow the Energy

Less than 50% of sunlight is **PAR**, and plants capture only 2–10% of it. Energy flows one way — it is **unidirectional**. At each step only about 10% passes on — **Lindeman**'s **10 percent law**.

💡 **REMEMBER**

The rest of the energy is lost as heat.



Food Chains

Grass → grasshopper → frog → snake → eagle is a **grazing food chain**. A **detritus food chain** starts with dead matter and decomposers; on land, more energy flows through it. Each step is a **trophic level**; the living mass at a level is its **standing crop**.

💡 **REMEMBER**

Food chains are interconnected to form a food web.



The Pyramids

Ecological pyramids show the levels. The **pyramid of energy** is always upright. The pyramid of numbers is inverted for a single big tree with many insects; the pyramid of biomass is inverted in the sea, where tiny phytoplankton support bigger fish.

💡 **REMEMBER**

Pyramids do not include saprophytes and treat each species as being at only one level.



The Changing Scene

Communities change over time — **ecological succession**. On bare rock, **lichens** come first as **pioneer species**, then mosses, herbs, shrubs and finally trees — the **climax community**. In water it starts with phytoplankton (hydrarch).

💡 **REMEMBER**

Secondary succession is faster because soil is already present.



Round and Round

Nutrients are recycled. The **carbon cycle** is a gaseous cycle; about 71% of carbon is dissolved in the oceans. The **phosphorus cycle** is a sedimentary cycle with rocks as its reservoir and no release by respiration. Bindu sighed happily: "Nature never wastes anything!"

💡 **REMEMBER**

Gaseous cycles have their reservoir in air/oceans; sedimentary cycles in Earth's crust.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

ecosystem p.1 · Producers p.1 · consumers p.1 · decomposers p.1 · stratification p.1 · gross primary productivity p.2 · net primary productivity p.2 · Decomposition p.3 · fragmentation p.3 · leaching p.3 · catabolism p.3 · humification p.3 · humus p.3 · mineralisation p.3 · PAR p.4 · unidirectional p.4 · Lindeman p.4 · 10 percent law p.4 · grazing food chain p.5 · detritus food chain p.5 · trophic level p.5 · standing crop p.5 · Ecological pyramids p.6 · pyramid of energy p.6 · ecological succession p.7 · lichens p.7 · pioneer species p.7 · climax community p.7 · carbon cycle p.8 · phosphorus cycle p.8



QUICK QUIZ

Do you remember?

1. NPP equals:

- GPP + R
- GPP − R
- R − GPP
- GPP × R

2. The 10 percent law was given by:

- Odum
- Lindeman
- Tansley
- Gause

3. Which pyramid is always upright?

- Number
- Biomass
- Energy
- All of these

4. Pioneer species on bare rock are usually:

- Mosses
- Lichens
- Grasses
- Trees

5. The reservoir of the phosphorus cycle is:

- Air
- Rocks
- Oceans
- Living organisms

Answers: 1-GPP − R · 2-Lindeman · 3-Energy · 4-Lichens · 5-Rocks